Unit 1

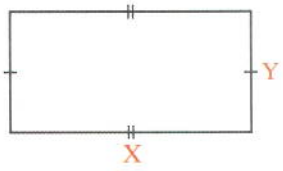
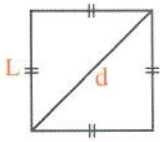
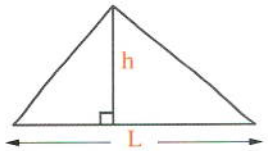
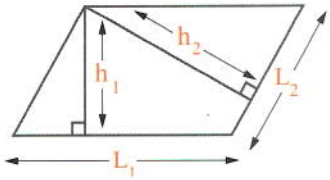
Lesson Four

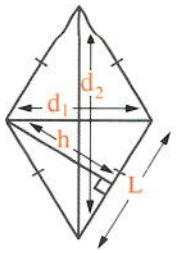
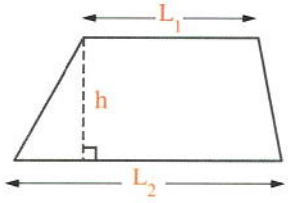
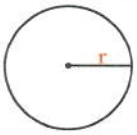
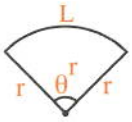
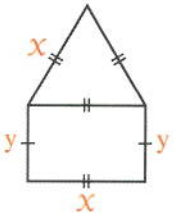
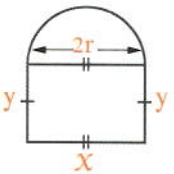
Related time rates

When a circular lamina is exposed to heat , then it expands , both its radius and its area expands and as long as it exposes to heat its radius and area changes.

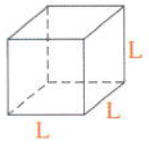
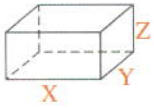
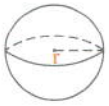
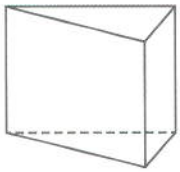
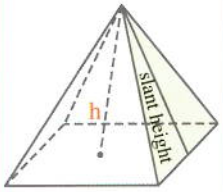
Let the radius length of the lamina be (r) and its area (A) , then $\frac{dr}{dt}$, $\frac{dA}{dt}$ (where t is the time) are called related time rates and from the relation between (r) and (A) the relation between $\frac{dr}{dt}$ and $\frac{dA}{dt}$ can be obtained , if one of them is given , the other can be calculated in general , if a relation between several variables x , y , z is given and by differentiating this relation with respect to time , relation between the related time rates $\frac{dx}{dt}$, $\frac{dy}{dt}$, $\frac{dz}{dt}$ can be found.

Remember area and perimeter of some geometrical figures :

Rectangle		Perimeter = 2 (x + y) Area = x × y
Square		Perimeter = 4 L Area = $L^2 = \frac{1}{2} d^2$
Triangle		Perimeter = sum of side lengths Area = $\frac{1}{2}$ × L × h = $\frac{1}{2}$ the product of any two sides × sine of included angle
Parallelogram		Perimeter = 2 ($L_1 + L_2$) Area = $L_1 \times h_1 = L_2 \times h_2$

Rhombus		Perimeter = 4 L Area = $L \times h$ $= \frac{1}{2} d_1 \times d_2$
Trapezium		Perimeter = sum of its side lengths Area = $\frac{1}{2} (L_1 + L_2) \times h$
Circle		Perimeter = $2 \pi r$ Area = πr^2
Sector		Perimeter = $2 r + L$ Area = $\frac{1}{2} L r = \frac{1}{2} \theta^{rad} r^2$ where $\theta^{rad} = \frac{L}{r}$, $\frac{\chi°}{180°} = \frac{\theta^{rad}}{\pi}$
Window in form of rectangle and equilateral triangle on its top		Perimeter = $3 \chi + 2 y$ Area = $\chi y + \frac{1}{2} \chi^2 \sin 60°$ $= \chi y + \frac{\sqrt{3}}{4} \chi^2$
Window in form of rectangle and semi-circle on its top		$r = \frac{1}{2} \chi$ Perimeter = $\chi + 2 y + \pi r$ $= \chi + 2 y + \frac{1}{2} \pi \chi$ Area = $\chi y + \frac{1}{2} \pi r^2$ $= \chi y + \frac{1}{8} \pi \chi^2$
Regular polygon	Where n is the number of sides and χ is the side length.	Perimeter = $n \chi$ Area = $\frac{1}{4} n \chi^2 \cot \frac{\pi}{n}$

Remember volume and lateral area and total surface area of some solid :

Solid		Lateral area	Total area	Volume
Cube		$4\,L^2$	$6\,L^2$	L^3
Cuboid		$2\,(x + y) \times z$	$2\,(x\,y + y\,z + z\,x)$	$x \times y \times z$
Right circular cylinder		$2\,\pi\,r\,h$	$2\,\pi\,r\,h + 2\,\pi\,r^2$ $= 2\,\pi\,r\,(h + r)$	$\pi\,r^2\,h$
Sphere		------	$4\,\pi\,r^2$	$\frac{4}{3}\,\pi\,r^3$
Right cone		$\pi\,r\,L$	$\pi\,r\,L + \pi\,r^2$	$\frac{1}{3}\,\pi\,r^2\,h$
Prism		Base perimeter × height	Lateral area + sum of areas of its bases	Area of its base × height
Regular pyramid		$\frac{1}{2}$ Base perimeter × slant height	Lateral area + area of its base	$\frac{1}{3}$ base area × height

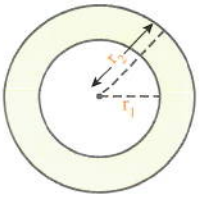
Remarks

1. If the variable $\mathcal{X}$ is
- Increasing / moving away / expanding / pouring into / accumulating
- Decreasing / approaching / shrinking / leaking / melting

→ as the time increase then $\frac{d\mathcal{X}}{dt}$ →
- positive
- negative

2. The distance between any two points : $(\mathcal{X}_1 , y_1)$, $(\mathcal{X}_2 , y_2)$ is $\sqrt{(\mathcal{X}_2-\mathcal{X}_1)^2+(y_2-y_1)^2}$

3. The volume of the bounded part between two concentric spheres , their radii lengths are $r_1 , r_2 = \frac{4}{3}\pi\left(r_2^3 - r_1^3\right)$



4. If $k = \mathcal{X} y z$ (Three variables) , then $\frac{dk}{dt} = \frac{d\mathcal{X}}{dt} \times y z + \frac{dy}{dt} \times \mathcal{X} z + \frac{dz}{dt} \times \mathcal{X} y$

5. The distance between the point $(\mathcal{X}_1 , y_1)$ and the straight line $a\mathcal{X} + b y + c = 0$ is $\frac{|a\mathcal{X}_1 + b y_1 + c|}{\sqrt{a^2+b^2}}$

Example 1

Point moves on the curve : $\mathcal{X}^2 + y^2 - 5\mathcal{X} + 4y + 9 = 0$ The rate of change of its $\mathcal{X}$-coordinate with respect to time at the point $(2 , -1)$ is 3, find the rate of change of its y-coordinate with respect to time at the same point.

Solution

$\because \mathcal{X}^2 + y^2 - 5\mathcal{X} + 4y + 9 = 0$ (by differentiating with respect to time)

$\therefore 2\mathcal{X}\frac{d\mathcal{X}}{dt} + 2y\frac{dy}{dt} - 5\frac{d\mathcal{X}}{dt} + 4\frac{dy}{dt} = 0$

$\therefore (2\mathcal{X} - 5)\frac{d\mathcal{X}}{dt} + (2y + 4)\frac{dy}{dt} = 0$

, $\because \frac{d\mathcal{X}}{dt} = 3$ at $(2 , -1)$

$\therefore (4 - 5) \times 3 + (-2 + 4)\frac{dy}{dt} = 0$

$\therefore -3 + 2\frac{dy}{dt} = 0$ $\quad\quad$ $\therefore \frac{dy}{dt} = \frac{3}{2}$

Example 2

Point moves on the curve : $x y = x + y - 5$, find the position of the point at which the rate of change of x-coordinate is equal to the rate of change of y-coordinate with respect to time.

Solution

$\because x y = x + y - 5$ (by differentiating with respect to time)

$\therefore x \frac{dy}{dt} + y \frac{dx}{dt} = \frac{dx}{dt} + \frac{dy}{dt}$, $\because \frac{dx}{dt} = \frac{dy}{dt}$

$\therefore (x + y) \frac{dy}{dt} = 2 \frac{dy}{dt}$ $\left(\text{divide by } \frac{dy}{dt}\right)$

$\therefore x + y = 2$ $\therefore y = 2 - x$

By substituting in the curve equation :

$\therefore x(2 - x) = x + 2 - x - 5$ $\therefore 2x - x^2 + 3 = 0$

$\therefore x^2 - 2x - 3 = 0$ $\therefore (x - 3)(x + 1) = 0$

$\therefore x = 3$ and so $y = -1$ $\therefore x = -1$ and so $y = 3$

$\therefore$ The points are : $(3 , -1) , (-1 , 3)$

Example 3

Gas is leaking out from a spherical balloon at rate 10π cm^3/sec. when the radius length = 10 cm. , find :

1. The rate of change of its radius length.

2. The rate of change of the balloon surface area.

Solution

Let the radius length of the balloon = r cm.

, the surface area of the balloon = A cm^2 , its volume = V cm^3

1. $\because V = \frac{4}{3} \pi r^3$ $\therefore \frac{dv}{dt} = 4 \pi r^2 \frac{dr}{dt}$

$\therefore -10\pi = 4\pi (10)^2 \times \frac{dr}{dt}$ $\therefore \frac{dr}{dt} = \frac{-1}{40}$ cm./sec.

2. $\because A = 4 \pi r^2$ $\therefore \frac{dA}{dt} = 8 \pi r \frac{dr}{dt}$

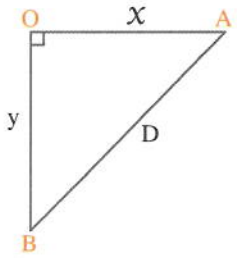
$\therefore \frac{dA}{dt} = 8\pi \times 10 \times \frac{-1}{40} = -2\pi$ cm^2/sec.

Example 4

A train starts its journey at 10 am. in direction of east with speed 20 km./h. At 12 pm. , another train moves from the same point towards south with speed 30 km./h. Find the rate of increase of the distance between them at 2 pm.

Solution

Let the first train at a distance ($\mathcal{X}$) km. from (O) and the second train at a distance (y) km. from (O) and the distance between the two trains (D) km.



$\therefore D^2 = \mathcal{X}^2 + y^2$

$\therefore 2D\frac{dD}{dt} = 2\mathcal{X}\frac{d\mathcal{X}}{dt} + 2y\frac{dy}{dt}$ (1)

At 2 pm : $\mathcal{X} = 4 \times 20 = 80$ km. , $y = 2 \times 30 = 60$ km.

$\therefore D = \sqrt{(80)^2 + (60)^2} = 100$ km.

From (1) : $\therefore 2 \times 100 \times \frac{dD}{dt} = 2 \times 80 \times 20 + 2 \times 60 \times 30$

$\therefore \frac{dD}{dt} = 34$ km./h.

Important Remark

Let $\mathcal{X}_o$ be the intial value of variable $\mathcal{X}_{(t=0)}$, and is the rate of change of $\mathcal{X}$ with respect to time is constant (**i.e.** $\frac{d\mathcal{X}}{dt}$ = constant value) then after time (t) the magnitude of the variable $\mathcal{X}$ is given by :

$$\mathcal{X} = \mathcal{X}_o + \frac{d\mathcal{X}}{dt} \times t$$

For example :

1. The length of a metal rod = 160 cm. is exposed to heat so that it expands by rate 2 cm./sec. , then the length of the rod after (t) sec.

$L = L_o + \frac{dL}{dt} \times t$

$\therefore L = 160 + 2t$

$L_o = 160$ cm.

$\frac{dL}{dt} = 2$ cm./sec.

2. A rocket of mass 17 ton is launched and it emits the fuel at rate 200 kg./sec.

, then the mass of the rocket after 40 seconds of the launching moment.

$m = m_o + \frac{dm}{dt} \times t$ | $m_o = 17$ ton $= 17000$ kg.

$m = 17000 - 200 \times 40 = 9000$ kg. $= 9$ ton | $\frac{dm}{dt} = -200$ kg./sec.

Example 5

The lengths of two sides of right-angled triangle are 8 cm. , 6 cm. , the length of the first side decreases by $\frac{1}{2}$ cm./min. and the length of the second increases by 1 cm./min.

Find :

1. The rate of change of surface area of the triangle after 2 minutes.

2. The time at which the increasing is vanished.

Solution

$\because$ The length of the first side is decreasing by rate $\frac{1}{2}$ cm./min.

$\therefore$ After t second its length $= \left(8 - \frac{1}{2}t\right)$ cm.

, the length of the second side is increasing by rate 1 cm./min.

$\therefore$ After t second its length $= (6 + t)$ cm.

Remember

$L = L_o + \frac{dL}{dt} \times t$

$\therefore$ The area of triangle (A) $= \frac{1}{2} \times \left(8 - \frac{1}{2}t\right) \times (6 + t)$

$= \left(4 - \frac{1}{4}t\right) \times (6 + t) = 24 + \frac{5}{2}t - \frac{1}{4}t^2$

$\therefore \frac{dA}{dt} = \frac{5}{2} - \frac{1}{2}t$ $\qquad$ $\therefore \left(\frac{dA}{dt}\right)_{t=2} = \frac{5}{2} - \frac{1}{2}(2) = 1.5$ cm^2/min.

$\therefore \frac{dA}{dt} = 0$ $\qquad$ $\therefore \frac{5}{2} - \frac{1}{2}t = 0$ $\qquad$ $\therefore t = 5$ min.

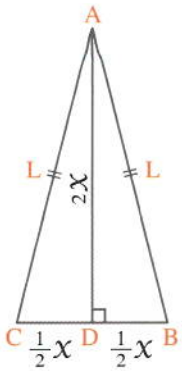
Example 6

A metal lamina in the form of an isosceles triangle its height = twice its base, the length of its base is increasing by heating by rate 0.04 cm. / sec. find :

1. The increasing rate of surface area of the lamina when its base 8 cm.

2. The rate of change of each side length.

Solution



Let the base length be X cm. $\therefore$ The length of its height = $2X$ cm.

, surface area of the triangle = A cm^2

and the length of each side = L

1. $\because A = \frac{1}{2} X \times 2X = X^2$ (by differentiation with respect to time)

$\therefore \frac{dA}{dt} = 2X \frac{dX}{dt}$

$\therefore \left(\frac{dA}{dt}\right)_{X=8} = 2 \times 8 \times 0.04 = 0.64$ cm^2/sec.

2. $\because L^2 = (2X)^2 + \left(\frac{1}{2} X\right)^2 = 4X^2 + \frac{1}{4} X^2 = \frac{17}{4} X^2$

$\therefore L = \frac{\sqrt{17}}{2} X$ (by differentiation with respect to time)

$\therefore \frac{dL}{dt} = \frac{\sqrt{17}}{2} \frac{dX}{dt}$ $\therefore \frac{dL}{dt} = \frac{\sqrt{17}}{2} \times 0.04 = 0.02\sqrt{17}$ cm./sec.

Example 7

The height of right-circular cylinder = $\frac{7}{6}$ of its base diameter length. Find the rate of change of its volume when the radius length of its base is 6 cm. , given that the rate of change of its height = 0.01 cm./sec.

Solution

Let the radius length of the base = r cm. and its height = h cm.

$\therefore h = \frac{7}{6} \times 2r$ $\therefore r = \frac{3}{7} h$

, $\because V = \pi r^2 h = \pi \left(\frac{3}{7} h\right)^2 \times h = \frac{9}{49} \pi h^3$ (by differentiating with respect to time)

$\therefore \frac{dv}{dt} = \frac{27}{49} \pi h^2 \times \frac{dh}{dt}$, at r = 6 cm.

$\therefore h = 14$ cm. $\therefore \frac{dv}{dt} = \frac{27}{49} \pi \times (14)^2 \times 0.01 = 1.08 \pi$ cm^3/sec.

Remarks

Let X be the radian measure of an angle then :

1. $\frac{d(\sin X)}{dt} = (\cos X) \times \frac{dX}{dt}$

2. $\frac{d(\tan X)}{dt} = (\sec^2 X) \times \frac{dX}{dt}$

3. $\frac{d(\cos X)}{dt} = (-\sin X) \times \frac{dX}{dt}$

From the previous remark , notice that :

$\odot \left[\frac{d(\sin X)}{dt}\right]_{X=0} = 1 \times \frac{dX}{dt}$, $\left[\frac{d(\tan X)}{dt}\right]_{X=0} = 1 \times \frac{dX}{dt}$

$\therefore \frac{d(\sin X)}{dt} = \frac{d(\tan X)}{dt}$ (at $X = 0$)

i.e. The sine and tangent of a given angle increase by the same rate at $(X = 0)$

$\odot \left[\frac{d(\cos X)}{dt}\right]_{X=\frac{\pi}{6}} = -\frac{1}{2} \times \frac{dX}{dt}$, $\left[\frac{d(\tan X)}{dt}\right]_{X=\frac{\pi}{6}} = \frac{4}{3} \times \frac{dX}{dt}$

$\therefore \frac{d(\cos X)}{dt} = \frac{-3}{8} \times \frac{d(\tan X)}{dt} \left(\text{at } X = \frac{\pi}{6}\right)$

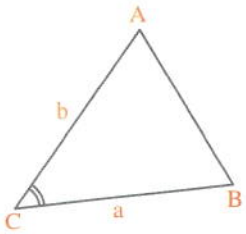
i.e. The rate of cosine any angle decreasing by rate $\frac{3}{8}$ as the rate of tangent the same angle increasing $\left(\text{at } X = \frac{\pi}{6}\right)$

Example 8

Two sides of a triangle , each increases by rate $\frac{1}{\sqrt{3}}$ cm. /sec. and the included angle increases by rate $\frac{1}{4}$ rad. /sec. Find the rate of change of surface area of the triangle when the length of each side = 12 cm.

Solution

Let the area of the triangle = A $\quad\quad$ $\therefore A = \frac{1}{2}\, a\, b \sin C$



$\therefore \frac{dA}{dt} = \frac{1}{2}\, a\, b \cos C \times \frac{dC}{dt} + \frac{1}{2} \sin C \left(a \frac{db}{dt} + b \frac{da}{dt}\right)$

, $\because \frac{da}{dt} = \frac{db}{dt} = \frac{1}{\sqrt{3}}$, $\frac{dC}{dt} = \frac{1}{4}$

, when the length of each side of the triangle = 12 cm. the triangle becomes an equilateral triangle.

$\therefore m(\angle C) = \frac{\pi}{3}$

$\therefore \frac{dA}{dt} = \frac{1}{2} \times 12 \times 12 \times \frac{1}{2} \times \frac{1}{4} + \frac{1}{2} \times \frac{\sqrt{3}}{2} \times \left(2 \times 12 \times \frac{1}{\sqrt{3}}\right) = 15 \text{ cm}^2/\text{sec.}$

Example 9

An isosceles triangle, the length of each side is 30 cm. and the measure of their included angle = χ, the rate of change of χ is 4 degree per minute. Find the rate of change of its surface area at $\chi = 60°$

Solution

$\because$ The area (A) $= \frac{1}{2} \times 30 \times 30 \times \sin \chi$

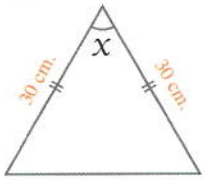
$\therefore A = 450 \sin \chi$ (by differentiating with respect to time)

$\therefore \frac{dA}{dt} = 450 \cos \chi \cdot \frac{d\chi}{dt}$

, at $\chi = 60°$

$\frac{d\chi}{dt} = 4° \times \frac{\pi}{180°} = \frac{\pi}{45}$ rad./min.

$\therefore \frac{dA}{dt} = 450 \cos 60° \times \frac{\pi}{45} = 5\pi$ cm^2/min.



Notice that

χ must be in radian.

Example 10

A 1.6 meter man walks away from a lamppost at a speed of 4 m./sec., the height of the lamppost is 4.8 m. from the ground. Find :

1. The rate of change of the length of the man's shadow.
2. The speed of end of the man's shadow.
3. The rate of change of the man's head from the lamp when the man is 2.4 m. from the lamppost.

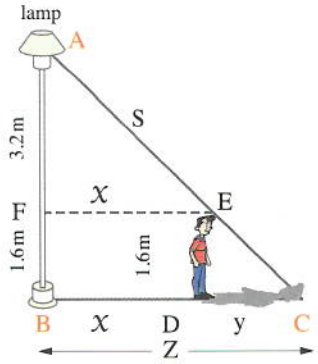
Solution

Let the distance between the man and the lamppost be χ

, the man's shadow = y

, the distance between the end of the man's shadow and the lamppost = z

, the distance between the man's head and the lamp = s



1. $\therefore \Delta ABC \sim \Delta EDC$

$\therefore \frac{4.8}{1.6} = \frac{\chi + y}{y}$

$\therefore 3y = \chi + y$

$\therefore 2y = \chi$ (Differentiate with respect to time)

$\therefore 2\frac{dy}{dt} = \frac{d\chi}{dt}$

$\therefore \frac{dy}{dt} = \frac{1}{2} \times 4 = 2$ m./sec.

2. $\because z = x + y$ $\quad\therefore \frac{dz}{dt} = \frac{dx}{dt} + \frac{dy}{dt} = 2 + 4 = 6$ m./sec.

3. $\because S^2 = x^2 + (3.2)^2$, at $x = 2.4$ m.

$\therefore S = 4$ m. (Differentiate with respect to time)

$\therefore 2S\frac{dS}{dt} = 2x\frac{dx}{dt}$ $\quad\therefore 2 \times 4 \times \frac{dS}{dt} = 2 \times 2.4 \times 4$

$\therefore \frac{dS}{dt} = 2.4$ m./sec.

Example 11

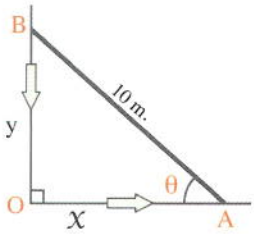
A 10 meter ladder is leaning by its lower end on a horizontal ground and its top on a vertical wall the lower end slides away from the wall at rate 2 m. /sec. When the lower end at a distance 8 m. from the wall. Find :

1. The rate of sliding the top of the ladder.

2. The rate of change of the measure of the angle between the ladder and the ground.

Solution

Let the lower end of the ladder at a distance x m. from the wall , the top end of the ladder at height y m. from the ground.



1. $\because x^2 + y^2 = 100$, at $x = 8$ m. , then $y = 6$ m.

$, 2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$ $\quad\therefore x\frac{dx}{dt} + y\frac{dy}{dt} = 0$

$\therefore 8 \times 2 + 6 \times \frac{dy}{dt} = 0$ $\quad\therefore \frac{dy}{dt} = \frac{-8}{3}$ m./sec.

2. $\because \sin\theta = \frac{y}{10}$ $\quad\therefore \cos\theta \cdot \frac{d\theta}{dt} = \frac{1}{10}\frac{dy}{dt}$

$\therefore \frac{8}{10} \times \frac{d\theta}{dt} = \frac{1}{10} \times \frac{-8}{3}$ $\quad\therefore \frac{d\theta}{dt} = -\frac{1}{3}$ rad./sec.

Example 12

Two intersecting roads , including angle of measure 120° , A car moves on one of them with speed 65 km./h. towards the point of intersection A

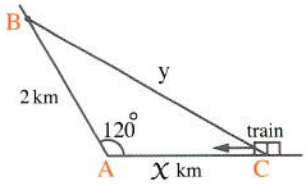
Find the rate of approaching of the car from station B which lies on the other road and at a distance 2 km. from A at the instant when the car is $1\frac{3}{4}$ km. from A.

Solution

Let the distance between A and the car be x km.

and the distance between the car and point B

in this instant = y km.



, $\because$ the car moves from A with speed 65 km./h.

$\therefore \frac{dx}{dt} = -65$ km./h.

From Δ ABC : $\because y^2 = x^2 + 2^2 - 2 \times x \times 2 \cos 120°$

$\therefore y^2 = x^2 + 2x + 4$ (Differentiate with respect to time)

$\therefore 2y \frac{dy}{dt} = (2x + 2) \frac{dx}{dt}$ $\therefore y \frac{dy}{dt} = (x + 1) \frac{dx}{dt}$ (1)

, $\because x = \frac{7}{4}$

$\therefore y^2 = \frac{49}{16} + \frac{7}{2} + 4 = \frac{169}{16}$ $\therefore y = \frac{13}{4} = 3\frac{1}{4}$ km.

By substitution in (1) :

$\therefore \frac{13}{4} \times \frac{dy}{dt} = \left(\frac{7}{4} + 1\right) \times -65$ $\therefore \frac{dy}{dt} = -55$

i.e. The rate of approaching of the car from point B at this moment = 55 km./h.

Example 13

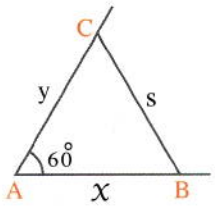
$\overrightarrow{AB}$, $\overrightarrow{AC}$ are two intersecting roads , including angle of measure 60°. A car moves on road $\overrightarrow{AB}$ from point A with speed 40 km./h. and after 10 minutes another car moves on the other road $\overrightarrow{AC}$ from point A with speed 60 km./h.

Find the rate of their divergence after 20 minutes from the beginning of motion of the second car.

Solution

Let the distance covered by the 1st car be x

$\therefore \frac{dx}{dt} = 40$ km./h.



and the distance covered by the 2nd car be y

$\frac{dy}{dt} = 60$ km./h.

, $\because S^2 = x^2 + y^2 - 2xy \cos 60°$

$\therefore S^2 = x^2 + y^2 - xy$ (1)

(Differentiate with respect to time)

$\therefore 2S\frac{ds}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt} - \left[x\frac{dy}{dt} + y\frac{dx}{dt}\right]$ (2)

and after 20 minutes from the beginning of motion of the 2nd car : $t = \frac{20}{60} = \frac{1}{3}$ h

$\therefore y = 60 \times \frac{1}{3} = 20$ km.

The time of the 1st car = (10 + 20 = 30 min) , so $t = \frac{30}{60} = \frac{1}{2}$ h.

$\therefore x = 40 \times \frac{1}{2} = 20$ km.

From (1) : $S = \sqrt{(20)^2 + (20)^2 - 20 \times 20} = 20$ km.

By substituting in (2) :

$\therefore 2 \times 20\frac{ds}{dt} = 2 \times 20 \times 40 + 2 \times 20 \times 60 - [20 \times 60 + 20 \times 40]$

$\therefore \frac{ds}{dt} = 50$ km./h.

Another Solution

$\because$ The covered distance by the 1st car before the 2nd car move $= 40 \times \frac{10}{60} = \frac{20}{3}$ km.

Assume the two cars move together time (t) h.

So , the covered distance by the 1st car $= 40t + \frac{20}{3}$

, the covered distance by the 2nd car $= 60t$

$\therefore S^2 = \left(40t + \frac{20}{3}\right)^2 + (60t)^2 - 2\left(40t + \frac{20}{3}\right)(60t) \times \frac{1}{2}$ (1)

, at t = 20 minutes $= \frac{20}{60} = \frac{1}{3}$ h.

$\therefore S = 20$ km.

by differentiating the relation (1) with respect to time (t)

$\therefore 2S\frac{ds}{dt} = 2\left(40t + \frac{20}{3}\right) \times 40 + 2(60t) \times 60 - \left[\left(40t + \frac{20}{3}\right)(60) + (60t) \times (40)\right]$

$\therefore 2S\frac{ds}{dt} = 5600t + \frac{400}{3}$

$\therefore 2 \times 20\frac{ds}{dt} = 5600 \times \frac{1}{3} + \frac{400}{3}$ $\therefore \frac{ds}{dt} = 50$ km./h.

Example 14

A rope of length 25 m. passes over a pulley at height 12 m. from the ground , tied at one of its ends by a mass and the other end by a car moves with speed 6 m./sec.

Find the rate of change of height of the mass at the moment that the car at a distance 16 m. from the projection of the pulley on the ground.

Solution

Let the height of the mass = h and the distance of the car from the projection of the pulley = x

$\therefore$ BC = 12 – h , AB = 25 – (12 – h) = 13 + h

From the figure :

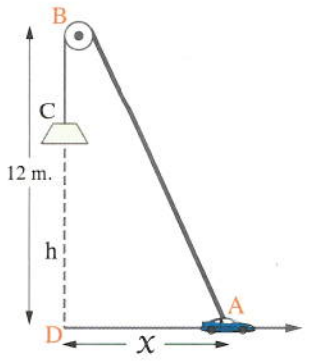
$\therefore (13 + h)^2 = x^2 + 12^2$ (1)

$\therefore 2\,(13 + h)\,\frac{dh}{dt} = 2\,x\,\frac{dx}{dt}$

at $x = 16$ m. , from equation (1) :

$\therefore (13 + h)^2 = 400$ $\quad\quad$ $\therefore h = 7$ m.

$\therefore 2\,(13 + 7)\,\frac{dh}{dt} = 2 \times 16 \times 6$ $\quad\quad$ $\therefore \frac{dh}{dt} = 4.8$ m./sec.



Example 15

$\overline{AB}$ is a rod of length 15 m. , rests with its lower end A on horizontal ground and at one of its point C on a vertical wall of height 9 m. the lower end A slides away from the wall at rate $\frac{1}{2}$ m. /min.

Find the rate of descending of the end B when it reaches the edge of the wall.

Solution

Let : AC = S , AD = x

$\therefore S^2 = x^2 + (9)^2$ $\quad\quad$ $\therefore 2\,S\,\frac{ds}{dt} = 2\,x\,\frac{dx}{dt}$

When the end of the rod reaches the edge of the wall.

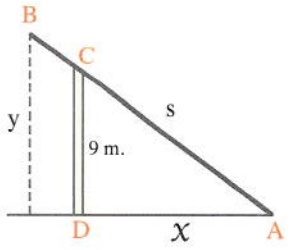
$\therefore S = 15$ m. , $x = 12$ m. $\quad\quad$ $\therefore 2 \times 15 \times \frac{ds}{dt} = 2 \times 12 \times \frac{1}{2}$

$\therefore \frac{ds}{dt} = 0.4$ m./min.

From the figure : $\frac{S}{15} = \frac{9}{y}$ $\quad\quad$ $\therefore S\,y = 135$

$\therefore S\,\frac{dy}{dt} + y\,\frac{ds}{dt} = 0$ $\quad\quad$ $\therefore 15\,\frac{dy}{dt} + 9 \times 0.4 = 0$

$\therefore \frac{dy}{dt} = -0.24$ m./min.



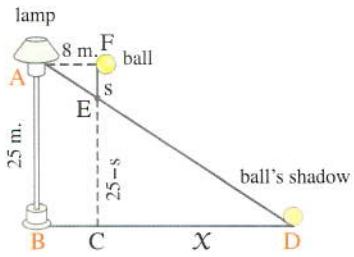
Example 16

A lamp fixed at the top of a tower , the tower height is 25 m. if a ball is fallen from the same height from a point 8 m. from the lamp. Find the speed of the ball's shadow on the ground when the ball is 15 m. high from the ground.

Solution

Let the ball descends (s) m after (t) sec.

and the shadow on the ground on this moment at a distance ($\mathcal{X}$) m from its projection on the ground (c)



$\because \Delta AFE \sim \Delta DCE$

$\because \frac{s}{25-s} = \frac{8}{\mathcal{X}}$

$\therefore \mathcal{X} = \frac{200-8s}{s} = \frac{200}{s} - 8$ (Differentiate with respect to t)

$\therefore \frac{d\mathcal{X}}{dt} = \frac{-200}{S^2} \times \frac{ds}{dt}$

At the moment the ball at 15 m. high S = 10 m.

, $\because V^2 + V_0^2 + 2gS$ $\therefore V^2 = 0 + 2 \times 9.8 \times 10 = 196$

$\therefore V = 14$ m./sec. $\therefore \frac{ds}{dt} = 14$ m./sec.

By substitution in (1) : $\therefore \frac{d\mathcal{X}}{dt} = \frac{-200}{(10)^2} \times 14 = -28$ m./sec.

i.e. The ball's shadow approch from its projection on the ground by speed 28 m./sec.